

Options Pricing

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Abstract

The Options Market is a market selling contracts known as options. These Options contracts give you the right to buy or sell a stock at a certain value within a certain expiration date. For American Options, you have the right to buy or sell a stock at any given time up to the expiration date. For European Options, you have the right to buy or sell a stock at the expiration date. What the right to buy or sell a stock means is that you are given the *Option* to buy or sell a stock without the obligation to. The price, known as the strike price, is a set value for the stock that the option gives you.

Maximizing profit is different for call and put options. For call options, you want the stock price to rise high above the strike price, so that you can buy the shares for less than they are being bought for on the stock market. For put options, you want the stock price to go low below the strike price so you can sell the shares for more than they are being sold for on the stock market. Your profit is also affected by the price you buy the contract for, which is determined by the bid and ask prices of the option. The bid price is the highest price a person is willing to buy the option for. The ask price is the lowest price a person is willing to sell the option for. When a buyer and a seller agree on a price, the option is sold for that price.

1 Introduction

There are many analytical methods to calculate your chances of profit on the options market, with one important one being the Black-Scholes Model. The Black-Scholes Model calculates the fair price of an options contract given its stock price, strike price, time to expiration in years, risk-free interest rate, and volatility. The Fair Price of an option is the expected value of an options contract under the assumptions of the Black-Scholes Model. If the fair price of an option is higher than its market price, that means the market¹ expects the option to be worth more, possibly because it is more volatile and has higher chances of profit. If the fair price of an option is less than its market price, that means the market expects the option to be priced below the market price, or that the option is overvalued and is potentially not worth its price.

However, Options don't necessarily have a "market price." Option's prices are determined by the bids and asks of the market, and until they meet, you cannot guess what the market price will be. Because of this, the market price is usually estimated. For this report, the market price will be estimated as:

$$MP = \frac{P_{\text{bid}} + P_{\text{ask}}}{2}$$

Which is the midpoint between the bid and ask prices. The distance between the bid and ask prices is known as the *spread*. We will define spread as:

$$\text{Spread} = P_{\text{ask}} - P_{\text{bid}}$$

Both Market Price and Spread allow us to see what the interaction between bid and ask is. A high spread tells you that the people trying to sell the contract value it much more than what people trying to buy the contract value it as. The market price tells you what the price the contract would be sold for would be if the bid and ask prices met in the middle, meaning this market price is *fair* for both the buyer and seller of the contract. We can use the market price and spread to also filter options. When looking at options contracts and determining what to buy, it is important to filter out any options that are clearly not good choices. For this project, the following terms will be passed for all options:

- Market Price must be above \$5.00
- Spread must be below \$5.00
- Volume must be nonzero
- Open Interest must be nonzero
- Bid Price must be nonzero

These conditions will assure that the options we are looking at are worth the inspection. Volume is the number of contracts traded per day, and open interest is the number of open contracts that have not been sold, expired, or closed.

2 Fair Price

Now, say we have a portfolio. What if we wanted to try and reduce the risk on this portfolio? We would want to do this so we can make the safest decisions on our finances.

2.1 Brownian Motion

Before looking at this risk-free portfolio. We must look at Geometric and Arithmetic Brownian Motion. Brownian Motion is a continuous-time stochastic process, which means it is defined using a random variable

¹The Market cannot truly expect something to happen, as it is not alive. This is based on mathematical expectation of what the data of the market shows. This will be explained further later, but expectation is defined through probability.

and independent over time intervals. This is very important in financial modeling, because stock price does not continuously change, it changes stochastically.

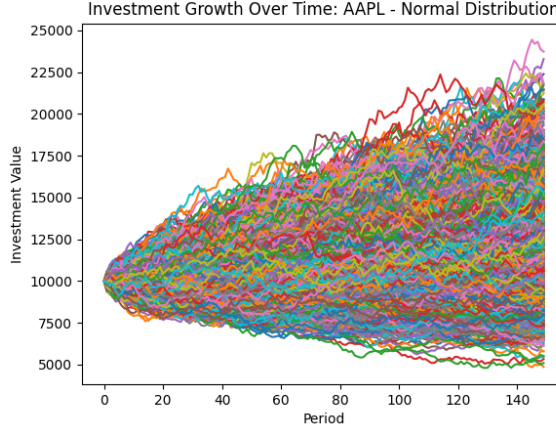


Figure 1: Enter Caption

This image is an example of a random walk predicting stock price using a normally distributed random variable.

Similar to Arithmetic and Geometric sequences and series, arithmetic brownian motion is additive, following a similar linear path, while geometric is multiplicative, following an exponential path.

The formulas for arithmetic and geometric brownian motion are defined as:

$$dX_t = \mu dt + \sigma dW_t \quad (\text{Arithmetic: 2.1})$$

$$dX_t = \mu X_t dt + \sigma X_t dW_t \quad (\text{Geometric: 2.2})$$

Intuitively, these make sense. Arithmetic Brownian Motion depicts a constant change μdt that has random shocks σdW_t applied to it. These shocks are what make Brownian Motion a stochastic process. For geometric brownian motion, the change is not constant, the value of the process directly affects the change in its value, with the once again random shocks. For this $dW_t = \epsilon \sqrt{\Delta t}$, where $\epsilon \sim N(0, 1)$. This is the random part of the whole equation.

An important thing to note about this process is that, brownian motion is continuous everywhere, but differentiable nowhere. This is because of the random shocks at every single value of t . To calculate this brownian motion, we approximate it using a certain number of steps over an interval, meaning the geometric equation² becomes

$$dX_t \approx X_t(\mu \Delta t + \sigma \epsilon \sqrt{\Delta t})$$

or

$$X_{i+1} = X_i(1 + \mu \Delta t + \sigma \epsilon \sqrt{\Delta t})$$

²For this, we will not focus on arithmetic brownian motion very much, because it is not as important for our purpose of options pricing. The arithmetic brownian motion equation is extremely similar, it is just missing the extra X_i term, so if you are interested you can look at that more closely.

Where X_{i+1} is the value of the next step when approximating the brownian motion, and Δt is the change in time between the steps.

I will not show the derivation, but from this geometric brownian motion we get a formula for the *fair price* of an option. This formula is called the Black-Scholes Equation.

$$C = SN(d_1) - Ke^{-rT}N(d_2) \quad (2.3)$$

$$P = Ke^{-rT}N(-d_2) - SN(-d_1) \quad (2.4)$$

Where C is the fair call price of an option, P is the fair put price, r is the risk free interest rate, T is the time to expiration, K is the strike price, $N(x)$ is the normal distribution cumulative density function, and d_1 and d_2 are given by the equations:

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + T\left(r + \frac{\sigma^2}{2}\right)}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Essentially, this fair price tells you the worth of an option in relation to all of its moving parts. This can be compared to the market price of an option. If the fair price is above the market price, this means that the option is worth more than it is trying to be sold for, or the market "expects" the option to be worth more than what people are asking for it.

3 The GUI

To accompany this Options Analysis, a GUI was coded to help display information in a better looking way. This GUI has 4 main parts, An options chain, a Volatility Smile, a Volatility Surface, and a Option Simulator. This GUI was coded using PySide6.

^SPX: Call			Stock Price: \$7718.6				Expiration Date: 2026-10-30			
Contract	Strike	Last Price	Bid	Ask	Change	Percent Change	Volume	Open Interest	Implied Volatility	Market Price
SPXW261030C03000000	3000.0	4555.91	0.0	0.0	0.0	0.0	0	0	1.0000000000000000	0.0
SPXW261030C04000000	4000.0	3690.37	3708.9	3718.9	0.0	0.0	1.0	0	0.620136618544888	3714.9
SPXW261030C05000000	5000.0	2735.31	2727.7	2727.7	35.54004	1.3103336	1.0	0	0.5468691203613283	2722.7
SPXW261030C06000000	6000.0	1876.0	1886.65	1847.7	1655.7	0.0	0.0	1.0	1.0000000000000000	1861.7
SPXW261030C07000000	7000.0	1048.62	1028.7	1028.7	-16.75	-0.8921643	1410.0	0	0.3853002369112305	1033.7
SPXW261030C08000000	8000.0	6025.0	1624.77	1550.9	1573.2	0.0	0.0	1.0	1.0000000000000000	1562.05
SPXW261030C09000000	9000.0	6105.0	1624.64	1605.5	1670.5	0.0	0.0	2.0	0.3738454229917407	1605.5
SPXW261030C10000000	10000.0	6226.0	1532.48	1607.5	1617.5	0.0	0.0	0.0	0.3800527762548828	1612.5
SPXW261030C11000000	11000.0	6300.0	1456.22	1433.5	1443.5	14.26001	0.88893243	2.0	0.34689221272853224	1438.5
SPXW261030C12000000	12000.0	6360.0	1413.79	1384.5	1384.5	0.0	0.0	18.0	0.33980349723815915	1389.5
SPXW261030C13000000	13000.0	6400.0	1322.55	1236.0	1346.5	0.0	0.0	1.0	0.3357881972792619	1343.0
SPXW261030C14000000	14000.0	6676.0	1133.11	1088.1	1078.1	0.0	0.0	0.0	0.28877205656374505	1073.1
SPXW261030C15000000	15000.0	6700.0	879.57	1048.2	1072.2	0.0	0.0	1.0	0.1801	1080.2
SPXW261030C16000000	16000.0	6726.0	911.87	917.0	0.0	0.0	0.0	2.0	1.0000000000000000	917.0
SPXW261030C17000000	17000.0	6760.0	1115.48	965.8	1006.8	0.0	0.0	1.0	0.2770343585561193	1000.8
SPXW261030C18000000	18000.0	6800.0	823.93	961.7	976.7	0.0	0.0	18.0	0.1839	963.7
SPXW261030C19000000	19000.0	6826.0	1036.35	923.2	933.2	0.0	0.0	4.0	0.28492663181104926	928.2
SPXW261030C20000000	20000.0	6860.0	796.71	903.7	923.7	0.0	0.0	1.0	0.1636	915.7
SPXW261030C21000000	21000.0	6880.0	823.93	870.4	893.4	0.0	0.0	1.0	0.2552158503860476	875.4
SPXW261030C22000000	22000.0	6900.0	857.45	861.5	861.5	0.0	0.0	2.0	0.2531917058187634	856.5
SPXW261030C23000000	23000.0	6915.0	674.29	0.0	0.0	0.0	0.0	1.0	1.0000000000000000	0.0
SPXW261030C24000000	24000.0	6920.0	660.52	832.0	855.0	0.0	0.0	10.0	0.1614	844.0
SPXW261030C25000000	25000.0	6925.0	826.91	824.2	844.2	0.0	0.0	1.0	0.1587	839.2
SPXW261030C26000000	26000.0	6960.0	798.64	784.3	804.3	0.0	0.0	1.0	0.1087	799.3
SPXW261030C27000000	27000.0	6980.0	607.17	0.0	0.0	0.0	0.0	1.0	1.0000000000000000	0.0
SPXW261030C28000000	28000.0	7000.0	778.32	756.7	766.7	12.39988	1.6097005	1410.0	0.1391	761.7
SPXW261030C29000000	29000.0	7020.0	883.17	793.5	773.5	0.0	0.0	2.0	0.1777	783.5
SPXW261030C30000000	30000.0	7020.0	748.03	738.0	747.4	0.0	0.0	1.0	0.1371	743.2
SPXW261030C31000000	31000.0	7025.0	676.08	0.0	0.0	0.0	0.0	2.0	1.0000000000000000	0.0
SPXW261030C32000000	32000.0	7040.0	953.0	883.4	703.3	0.0	0.0	4.0	0.16337503451338064	887.35
SPXW261030C33000000	33000.0	7050.0	783.57	712.7	733.8	0.0	0.0	1.0	0.1469	716.3
SPXW261030C34000000	34000.0	7080.0	617.19	678.3	682.8	0.0	0.0	18.0	0.182221007468154787	679.55
SPXW261030C35000000	35000.0	7085.0	737.62	692.7	706.5	0.0	0.0	2.0	0.146	702.1
SPXW261030C36000000	36000.0	7070.0	626.45	657.7	676.6	0.0	0.0	5.0	0.185885368041992	671.65

Figure 2: Options Chain for S&P500

3.1 Options Chain

The options chain displays all valuable information for all options, filtering out unnecessary ones, such as options with 0 volume and extremely low market prices compared to the stock price.

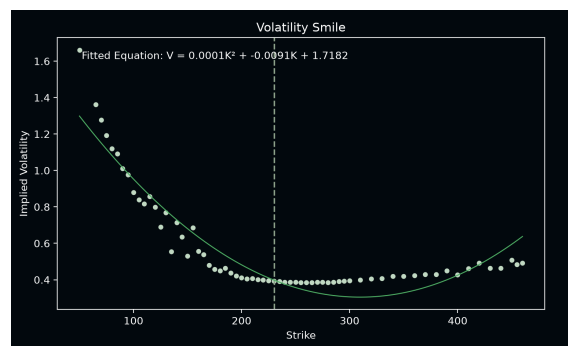
The options chain is mostly only market data pulled from online. But this market data helps make extremely informed decisions and use other tools to further analyze options more closely. This is not a simulation, but this information will be the main data used to make predictions and decisions.

3.2 The Volatility Smile

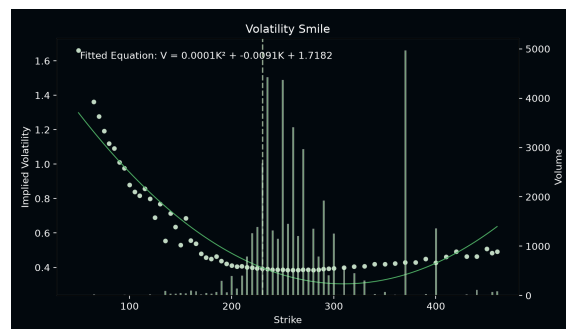
Volatility is often extremely difficult to calculate accurately for a stock. While volatility can be calculated as the standard deviation of returns, this method comes with assuming constant volatility. Assuming volatility is constant and calculated by standard deviation does not account for skew and fat tails generally found in the stock market. While this method of volatility calculation does assume constant volatility up to the expiration date, we can use the Black-Scholes Equation to reverse calculate volatility. This volatility is not historical, instead, it can be used to interpret an expectation of future volatility, and is therefore called *Implied Volatility*. It is important to note that implied volatility and historical volatility are not necessarily the same thing, even past the fact that one is forward looking and one is backwards looking, they measure different things.

Implied Volatility tells us how much the stock is predicted to move based off of the inputs into the Black-Scholes Equation, letting us know how the stock price is predicted to change. A higher implied volatility means that the stock price will shift more drastically. When using geometric brownian motion to predict stock price, a higher implied volatility will create more drastic shocks.

For this project, An implied volatility smile and implied volatility surface were created. An implied volatility smile is a plot of all implied volatilities for a certain expiration date against its strike price. Volatility smiles are not always smile shaped. They could have a slight skew or large curvature, but the main similarity is that they are nonlinear, and therefore a quadratic regression is calculated along with the smile. Open Interest and Volume Histograms are calculated as well for each strike price and displayed behind the volatility smile.



(a) NVDA Volatility Smile and Regression



(b) NVDA Volatility Smile, Regression, and Volume

Figure 3: NVDA Volatility Smiles with and without Volume Histogram

The volatility smile helps show how implied volatility changes as strike price changes. For many equity options, implied volatility varies systematically with strike price, often producing a downward-sloping skew rather than a symmetric smile. It is difficult to interpret these changing implied volatilities. Implied volatility measures how much the underlying stock price is expected to change in the future, so which one do you choose if you have multiple implied volatilities? This is a problem that comes with the realistic data, because the mathematical black-scholes model assumes constant volatility.

Implied Volatility will be analyzed more later on in this report. For now, the volume and implied volatility plot helps show what implied volatility and strike price the public is more interested in.

3.3 The Monte Carlo Simulation

This project allows the user to pick any option from the chain and run a monte carlo simulation on it. This simulation uses geometric brownian motion, where μ is the risk-free interest rate, and σ is the implied volatility. From this, a histogram of profits is calculated. Statistics for Profit and Payoff are calculated too, where payoff is defined as: $\max(S - K, 0)$ for calls and $\max(K - S, 0)$ for puts. It is important to restrict payoff to only positive values. Suppose you buy an call option and then its stock price goes below the strike price. Options do not give you an obligation to buy the stock at the strike price, so you can just ignore the option, making your only loss the price you paid for the contract. The same goes for put options, making profit for Call options: $\max(S - K, 0) - MP$ and profit for put options: $\max(K - S, 0) - MP$

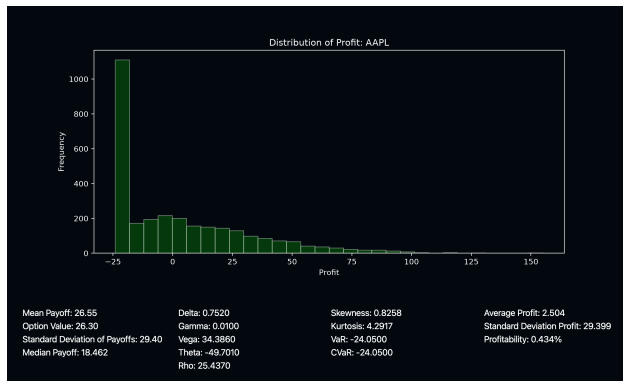


Figure 4: Profit Distribution: AAPL

Note the extremely high frequency around a -25 profit. This comes from the definition of payoff cutting payoff to only nonnegative values. This means that the extremely high frequency includes all final stock prices below and at the strike price.

The statistics calculated were the average and standard deviation payoffs, the option value, the median payoff, the greeks, skewness, kurtosis, CVaR, VaR, Profitability, Average Profit and Standard Deviation of Profit.

Some of these statistics are more important than others. For example, the payoff statistics could lead you to false conclusions, because they don't account for the price you pay for the option like the profit does. Because of that, we won't look very much at payoff statistics, only profit statistics. Not only that, but the standard deviation of profit and payoff are the same, because the only change in the data is a horizontal shift left by the market price.

The Option Value is a fairly important statistic. This value comes from the average payoff, but it accounts for the interest rate and time to expiration.

$$\bar{C} = X e^{rT}, \quad X = \bar{C} e^{-rT}$$

Where $\bar{C}$ is the average payoff, and X is the average payoff in terms of today's currency and market value. The reason this is important is because it gives you the *value* of an option, by its average or expected payoff, adjusted to what its value today would be. This is extremely important because you can use this to gauge where the market value will be and whether or not that market value is worth paying for.

Another important statistic is the *profitability*, which is defined as:

$$\frac{|P_+|}{|P|} \quad P_+ = \{p \in P | p > 0\}$$

Where P is the set of profits. This means profitability is the proportion of simulations that made a profit. It is important to note that because the geometric brownian motion used assumes risk-free interest rate is the average, this simulation is under risk-neutral measure, and proportions should not be interpreted as the realistic probability of profit.

3.3.1 The Greeks

The final extremely important statistic provided is the greeks. The greeks shown are Delta, Gamma, Vega, Theta, and Rho, which are defined as³:

$$\Delta = \frac{\partial C}{\partial S} \quad \Gamma = \frac{\partial^2 C}{\partial S^2} \quad \Theta = \frac{\partial C}{\partial T} \quad \nu = \frac{\partial C}{\partial \sigma} \quad \rho = \frac{\partial C}{\partial r}$$

And we can define $\nabla C = (\Delta, \frac{\partial C}{\partial K}, \Theta, \nu, \rho)$.

4 Implied Volatility

A stock has 1 volatility, however every single option has a different implied volatility, so which implied volatility do we interpret as the volatility? First let's define how we found this implied volatility.

The Black-Scholes equation cannot be solved for implied volatility, Because there is no algebraic way to do this. The Black-Scholes equation is transcendental. We also do not have the exact option value produced by the equation. So instead, a numerical method will be used to calculate the implied volatility for a certain option. The basic idea is: Choose a first guess implied volatility, calculate the Black-Scholes fair market price using that guess, and compare this to the market price of the option. Using this comparison, adjust the volatility guess accordingly. Next, this is repeated using the new adjusted implied volatility until the implied volatility returns a fair market price equal to the market price within a certain degree. This final iteration is the implied volatility.

4.1 The Newton-Raphson Method

The Newton-Raphson method is a process for finding the roots of functions by approximation. The formula comes simply from the formula for a line:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

We can swap out the x and $f(x)$ out for implied volatility and market price making the equation:

$$\sigma_{i+1} = \sigma_i - \frac{C(\sigma_i) - C_{market}}{\nu(\sigma_i)}$$

The reason why ν is used is because option price becomes a function of only volatility because everything else is constant, making the derivative the partial derivative with respect to volatility, ν . Not only that, but the reason for the C_{market} is because the function becomes $f(\sigma) = C(\sigma) - C_{market}$, to account for the root solving aspect of the Newton-Raphson method.

This is one method of approximating implied volatility, which was done until the implied volatility creates a difference of less than 10^{-5} between the market price and calculated fair market value. A downside of this method is that it can sometimes produce confusing results if the function has unfavorable local behavior. The black-scholes equation is a very nonlinear, but if the guess is close enough the Newton-Raphson method works very well. However, if the guess is not close enough then the Newton-Raphson method will provide an error.

³Or as a function of P , put price if you are analyzing Put Options.

As an example, suppose you are trying to find the zeros of $f(x) = (x - 1)(x + 1)$. This has a minimum at $x = 0$, and this will be the starting point. $f'(x_0) = 0$, so this immediately gives an error. While with a simple example like this it does not matter, with a complex function like the black-scholes equation this will not work the best.

4.2 Brent's Method

Implied volatility is the value of σ that causes the Black-Scholes model price to equal the observed market price of an option. Since the Black-Scholes pricing formula does not have a closed-form inverse with respect to volatility, a numerical root-finding method is required.

For a call option, define the function

$$f(\sigma) = C_{\text{BS}}(S, K, T, r, \sigma) - C_{\text{market}},$$

where C_{BS} is the Black-Scholes call price and C_{market} is the observed market price. The implied volatility is therefore the solution to

$$f(\sigma) = 0.$$

Similarly, for a put option,

$$f(\sigma) = P_{\text{BS}}(S, K, T, r, \sigma) - P_{\text{market}}.$$

4.2.1 Root Bracketing

Brent's method begins with two volatility values, σ_a and σ_b , such that

$$f(\sigma_a)f(\sigma_b) < 0.$$

By the Intermediate Value Theorem, if f is continuous, there must exist at least one root in the interval

$$[\sigma_a, \sigma_b].$$

For example, one might initially choose

$$\sigma_a = 0.01, \quad \sigma_b = 5.00.$$

If

$$f(0.01) < 0 \quad \text{and} \quad f(5.00) > 0,$$

then the implied volatility is known to lie somewhere between these two values.

Brent's method is a hybrid root-finding algorithm that combines three methods: bisection, the secant method, and inverse quadratic interpolation.

Bisection uses the midpoint of the current interval,

$$\sigma_m = \frac{\sigma_a + \sigma_b}{2},$$

and evaluates $f(\sigma_m)$. The endpoint that has the same sign as $f(\sigma_m)$ is then replaced, preserving the bracket containing the root.

The secant method instead approximates the function using a line through two known points. Given $(\sigma_a, f(\sigma_a))$ and $(\sigma_b, f(\sigma_b))$, the next approximation is

$$\sigma_s = \sigma_b - f(\sigma_b) \frac{\sigma_b - \sigma_a}{f(\sigma_b) - f(\sigma_a)}.$$

Brent's method can also use inverse quadratic interpolation. Given three points $(\sigma_a, f(\sigma_a))$, $(\sigma_b, f(\sigma_b))$, and $(\sigma_c, f(\sigma_c))$, a quadratic approximation to the inverse function is constructed to estimate the value of σ for which $f(\sigma) = 0$.

The key advantage of Brent's method is that it does not rely exclusively on any one of these methods. When interpolation produces a reliable estimate, Brent's method can converge much faster than bisection. If an interpolation step becomes unreliable, the algorithm falls back toward a bisection step, maintaining the root bracket.

For an option with market price V_{market} , the algorithm can therefore be summarized as follows:

$$f(\sigma) = V_{\text{BS}}(\sigma) - V_{\text{market}}.$$

Choose an initial bracket

$$[\sigma_a, \sigma_b]$$

satisfying

$$f(\sigma_a)f(\sigma_b) < 0.$$

Brent's method then repeatedly produces a new estimate σ_n and evaluates

$$f(\sigma_n).$$

The interval is updated so that the root remains bracketed. The process continues until the pricing error satisfies a specified tolerance,

$$|V_{\text{BS}}(\sigma_n) - V_{\text{market}}| < \varepsilon.$$

The resulting value

$$\boxed{\sigma_{\text{IV}} \approx \sigma_n}$$

is the implied volatility of the option.

Unlike Newton's method, Brent's method does not require the calculation of Vega or any other derivative of the Black-Scholes pricing function. Its requirement of a valid initial bracket also provides a useful guarantee that the algorithm will maintain an interval containing a root, provided the pricing function is continuous and the initial bracket contains a sign change.

Implied Volatility does still risk divergence at deep out-of-the-money and near expiration options. Some options make it extremely difficult to accurately calculate implied volatility, making it so even if a good numerical method is found to approximate it, it might be somewhat pointless to interpret or make decisions based on.

4.3 Implied Volatility and Historical Volatility

Implied Volatility interprets what the volatility of the stock would need to be to make the option's market value equal to its market price. This interpretation is extremely different from historical volatility, which measures market change and risk based off of historical shifts. The best way to interpret implied volatility is to compare it with other implied volatilities.

If an implied volatility is higher than another, the market expects more change and shock to make that option's market value the market price.

Suppose $\sigma_{\text{historical}} < \sigma_{\text{implied}}$. This means that the historical market does not have enough deviation to make the option's market value equal to its market price. When inputting the historical volatility into the black-scholes equation, a lower historical volatility would produce an option value below the market price, while a higher historical volatility would produce an option value above the market price. An option value above the market price does not necessarily mean an option worth purchasing though. This higher historical volatility could be too risky to buy in some cases.

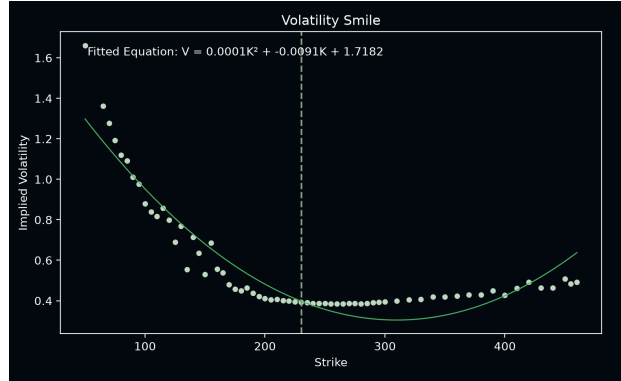


Figure 5: NVDA Implied Volatilities

4.4 The Volatility Surface

A way to visualize changes in implied volatility with respect to time to expiration and strike price is the volatility surface. This is a 3-Dimensional surface is a plot of implied volatility at every strike price and expiration date for a certain company.

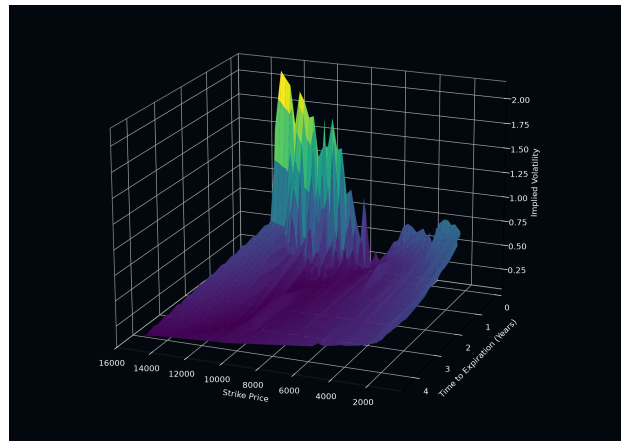


Figure 6: S&P500 Volatility Surface

There are many things to point out about this surface. Firstly, the surface is extremely jagged, due to the fact that implied volatility is being approximated at every value, and some options are highly illiquid and cause extremely non-smooth implied volatilities. Not only that, but the extremely high implied volatilities are most likely extremely out of the money options, creating extremely high implied volatilities.

A horizontal plane of historical volatility, or knowing the historical volatility would be useful for seeing what kind of options are above the historical volatility, and what kind are below. Note that the peak in implied volatility also occurs extremely close to the expiration date, and at very high strike prices, while options with low strike prices and large times to expiration have extremely low implied volatilities.

5 Optimizing Option Value

5.1 Call Options

Define the function:

$$C(S, K, T, r, \sigma) = SN(d_1) - Ke^{-rT}N(d_2)$$

with d_1 and d_2 defined as before. Optimization can be used to determine a maximum option value, within certain bounds. It is illogical to use every possible value for all inputs, because inputs cannot usually tend towards infinity or zero cleanly. Instead, define:

$$D : \{(S, K, T, r, \sigma) \in \mathbb{R}^5 | S_{min} \leq S \leq S_{max}, K_{min} \leq K \leq K_{max}, T_{min} \leq T \leq T_{max}, r_{min} \leq r \leq r_{max}, \sigma_{min} \leq \sigma \leq \sigma_{max}\}$$

Where all maximum and minimum values are positive real numbers. The goal is to find a maximum on D . Because The Black-Scholes function is continuous, the extreme value theorem guarantees a maximum on D . Using the partial derivatives,

$$\nabla C = 0$$

However, an interior extrema requires $\nabla C = 0$, which is impossible with the black-scholes equation. Because of this, the maximum option value will exist on the boundary.

This is extremely hard to compute by hand using general methods in multivariable calculus. A 5 dimensional hyperbox has too many components that need to be computed. Instead, we can take advantage of the monotonicity of the partial derivatives of the Black-Scholes equations.

Theorem 1. *The Theoretical Maximum Option Value for a Call Option is:*

$$(S_{max}, K_{min}, T_{min}, r_{max}, \sigma_{max})$$

Proof.

$$\nabla C = (\Delta, \frac{\partial C}{\partial K}, \Theta, \rho, \nu)$$

$\Delta = N(d_1)$, Cumulative distribution functions are defined as $F(x) = \int_{-\infty}^x f(x)dx$, where $f(x)$ is a probability density function. Probability density functions must be always positive, meaning the cumulative distribution function is always increasing. Therefore, Δ is always positive and $C(S, K, T, r, \sigma)$ is strictly increasing in the delta direction. This means the highest option value only according to stock price will occur at S_{max}

Strike price is not continuous like stock price, time, interest rate, or volatility. However, for this strike price will be assumed continuous, so that a calculation of the partial derivative works.

$$\frac{\partial C}{\partial K} = \frac{\partial}{\partial K}(SN(d_1) - Ke^{-rT}N(d_2))$$

By the chain and product rules,

$$\frac{\partial C}{\partial K} = S \frac{\partial N}{\partial d_1} \frac{\partial d_1}{\partial K} - e^{-rT}N(d_2) - Ke^{-rT} \frac{\partial N}{\partial d_2} \frac{\partial d_2}{\partial K}$$

$$\frac{\partial N}{\partial x}(x) = \phi(x), \text{ The normal distribution probability density function, and}$$

$$\frac{\partial d_1}{\partial K} = \frac{\partial d_2}{\partial K}, \text{ Because } d_2 = d_1 - \sigma\sqrt{T}.$$

$$\frac{\partial C}{\partial K} = S\phi(d_1) \frac{\partial d_1}{\partial K} - e^{-rT}N(d_2) - Ke^{-rT} \phi(d_2) \frac{\partial d_1}{\partial K}$$

There exists a substitution such that $S\phi(d_1) = Ke^{-rT}\phi(d_2)$ making the equation:

$$\frac{\partial C}{\partial K} = Ke^{-rT}\phi(d_2) - e^{-rT}N(d_2) - Ke^{-rT}\phi(d_2)$$

$$\frac{\partial C}{\partial K} = -e^{-rT}N(d_2)$$

This is strictly decreasing, because both the exponential and cumulative distribution function term are always positive, meaning that the maximum option value occurs at K_{min} .

$$\Theta = -\frac{S\Phi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$$

This is negative, so the option value is strictly decreasing, meaning the maximum option value occurs at T_{min} .

$$\rho = KTe^{-\frac{r}{T}}N(d_2)$$

This is positive, so option value is strictly increasing, meaning the maximum option value is at r_{max} .

$$\nu = S\sqrt{T}\phi(d_1)$$

This is always positive, so option value is strictly increasing, meaning the maximum option value is at σ_{max} .

These values are where the maximums occur, meaning the maximum option value occurs at

$$(S_{max}, K_{min}, T_{min}, r_{max}, \sigma_{max}).$$

□

5.2 Put Options

Define the function:

$$P(S, K, T, r, \sigma) = Ke^{-rT}N(-d_2) - SN(-d_1)$$

Once again, this function can be optimized under the same conditions as before.

Theorem 2. *The Theoretical Maximum Option Value for a put option is:*

$$(S_{min}, K_{max}, T, r_{min}, \sigma_{max}),$$

Where T varies based upon bounds and must be found using numerical approximation methods.

Proof.

$$\nabla P = (\Delta, \frac{\partial C}{\partial K}, \Theta, \rho, \nu)$$

For $\frac{\partial P}{\partial S}$, $\Delta = N(d_1) - 1$, because cumulative distribution functions are always between 0 and 1, Δ is always negative, meaning the option's value is strictly decreasing, so the maximum option value occurs at S_{min} .

For $\frac{\partial P}{\partial K}$,

$$\frac{\partial P}{\partial K} = \frac{\partial}{\partial K}(Ke^{-rT}N(-d_2) - SN(-d_1))$$

By the chain rule and product rule,

$$\frac{\partial P}{\partial K} = e^{-rT}N(-d_2) - Ke^{-rT}\frac{\partial N}{\partial d_2}\frac{\partial d_2}{\partial K} + S\frac{\partial N}{\partial d_1}\frac{\partial d_1}{\partial K}$$

$$\frac{\partial d_1}{\partial K} = \frac{\partial d_2}{\partial K}, \text{ and } \phi(x) = \frac{\partial N}{\partial x}$$

$$\frac{\partial P}{\partial K} = e^{-rT} N(-d_2) + \frac{\partial d_1}{\partial K} (Ke^{-rT} \phi(-d_2) - S\phi(-d_1))$$

$$S\Phi(-d_1) = Ke^{-rT} \phi(-d_2), \text{ meaning}$$

$$\frac{\partial P}{\partial K} = e^{-rT} N(-d_2)$$

This is always positive, meaning option value is strictly increasing, and the maximum option value occurs at K_{max} .

For $\frac{\partial P}{\partial r}$, $\rho = -Ke^{-rT} N(d_2)$. This is always negative, meaning option value is strictly decreasing, and the maximum occurs at r_{min} .

For $\frac{\partial P}{\partial \sigma}$, ν is the same for call and put options, meaning option value is strictly increasing, making the maximum option value occur at σ_{max} .

For $\frac{\partial P}{\partial T}$,

$$\Theta = -\frac{S\phi(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT} N(-d_2)$$

Unlike other inputs, the time to expiration for put options is not monotonic. This means that the maximum option value could be found at either end of the boundary or inside the boundary. Furthermore, the equation for Θ is transcendental because there is a time variable as an integral bound and in an exponential term. This means that $\Theta = c \forall c \in \mathbb{R}$ cannot be solved using algebraic techniques. Therefore, this requires a numerical approximation to find the maximum, which varies based on values chosen for minimum and maximum stock price, strike price, interest rate, time to expiration, and volatility.

This makes the maximum option value occur at $(S_{min}, K_{max}, T, r_{min}, \sigma_{max})$, Where T can be found using numerical methods, such as the Newton-Raphson method. $\square$

These 2 maximums of option value could be used to gauge what options are best to look into. However, this is an ideal theoretical result. There will not always exist an option such that it satisfies

$(S_{max}, K_{min}, T_{min}, r_{max}, \sigma_{max})$ compared to all other options. First of all, the stock price depends entirely on the market, which is difficult to predict.

Suppose one finds an option with the highest value, this option will have an extremely high market price, often making it so there is no or negative profit. As an example, the option $O_{AAPL}(260, 2)$ (260 strike price, 2 days to expiration) is the theoretical highest option value for AAPL, with a value of \$74.78. However this does not necessarily mean this is the option everyone should be buying. The profitability is 0.501%. This is not the best sounding for an option, and the Conditional Value at Risk is -70 as well. Under the assumptions of the simulation, the option is not attractive to a buyer.

6 Limitations of The Black-Scholes Model

There are a few problems with the Black-Scholes model that cause it to lose accuracy in certain places. The Black-Scholes model simplifies some parts of the market, making its behavior not match market behavior. One of the biggest problems is the assumption that volatility is constant over the entire time until the option is exercised. Volatility changes continuously as new information on stocks continues to come out and the stock changes over and over again. This constant volatility ignores shocks that could happen due to extreme market changes. Suppose volatility was describes as a function $\sigma(S, T, r)$. For the black-scholes model to become more accurate, this volatility could be modeled using a stochastic process.

Volatility generally has 2 main patterns. Volatility clusters, meaning high volatility time periods group together. For example, high volatility days group themselves with other days, instead of just seeing a single high volatility peak every once in awhile. Volatility also experiences mean reversion. A volatility extreme due to the market will often drift back down to the average volatility. Having a volatility function that accounts for these shocks and clusters would make the black-scholes model much more accurate. Some more accurate ways to model volatility would be using GARCH, or Generalized Autoregressive Conditional Heteroskedasticity. This model measures variance more accurately to market variance, accounting for risk even more.

GARCH takes in two parameters, writing the model as GARCH(p,q). The simplest form of the equation is GARCH(1,1). The equation has multiple components: $y_t = \mu + \varepsilon_t$, where y_t is the asset's return at time t, μ is the expected return, and ε_t is the unexpected return residual, defined as $\varepsilon_t = \sigma_t z_t$. Where $z \sim N(0, 1)$, the randomly distributed part of the equation, and σ_t is the conditional volatility.

The main accuracy part of the GARCH model comes from the definition $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$. This shows that GARCH follows a recursive definition, using the previous period's conditions to generate the current one. This allows for volatility to be affected by clustering and shocks. ω is a baseline constant variance, α is the weight given to last time period's shock ε , and β is the weight to last time period's generated variance. Together this equation produces a variance that is more accurate than the constant Black-Scholes variance. The GARCH model has certain parameters for ω, α , and β also. $\omega > 0$, $\alpha, \beta \geq 0$, and $\alpha + \beta < 1$.

The Black-Scholes Model also assumes constant interest rates. This does not matter very much for short-term options, but for long-term options, such as expiration dates 2 years in the future, their interest rates change. The Black-Scholes model is an ideal model of the financial market, taking out changing volatility, interest rates, and fees that come with buying and selling options. Dividends are ignored, and returns are assumed lognormal. The lognormal distribution assumes that returns can still be somewhat predicted in a distribution, which ignores extreme crashes and is an optimistic view of the financial market.

The Black-Scholes Model is only applicable to European Options as well. Other Vanilla and Exotic options need other models to simulate and analyze them. American Options, options that can be exercised any time up to the expiration date, use the Binomial model to analyze them. The binomial model uses a discrete number of steps with an up state and a down state to calculate the possibilities and finds the option value by working backwards from the final state. The model is known as the binomial model because every step has an up and a down state, or 2 states for each step. These steps compounded together link to a binomial distribution. The binomial model can handle changing volatility unlike the black-scholes model, and it also takes in account complex payments due to American Options.

The Black-Scholes model has many limitations in the modern market, but it is still extremely useful for pricing european options. The Black-Scholes model is not only useful for its fair option value either. The greeks, implied volatility, and other properties make the Black-Scholes model a powerful prediction and pricing tool.